Exploring the Role of Economic Integration in Inequality: A Fixed Effects and First Difference Analysis of Trade Openness and the Gini Index

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Abstract

This paper employs fixed effects and first difference models to examine the effect of economic integration, represented by trade openness, on inequality. Panel data from the World Bank DataBank is used to look at the short- and long-term relationship between 1996 and 2019. Using the small set of countries with complete data, fixed effects and first difference estimates are computed, taking into account time-invariant country characteristics. Due to concerns of heteroscedasticity and serial correlation, inference is conducted through bootstrapping and clustering at the country level. While the resulting estimates are positive, they are practically and statistically insignificant. Thus it is suggested that the effect of trade openness on the Gini index is not different from 0. Potential bias may be a concern as a result of the small number of units and time-varying country characteristics that are not accounted for in the models.

I Introduction

The recent threats and implementations of tariffs have ignited larger conversations around the potential effects of these measures – typically regarding the impact on employment, prices, investment, and industries. As recently observed, these tariffs are often accompanied by retaliatory measures, affecting multiple parties and reducing overall economic integration. Exploring the impact of economic integration on societies within countries is therefore a relevant and timely area of study. In particular, inequality, often measured by the Gini index, serves as a key measure of societal conditions. Thus this paper will explore the impact of trade openness, used to represent economic integration, on the Gini index.

In doing so, this paper employs fixed effects and first difference models to account for time-invariant country specific characteristics that may affect both trade openness and inequality. Panel data is pulled from the World Bank DataBank, containing information on trade openness, the Gini index, and other control variables for countries between 1996 and 2019. Possible bias is introduced due to the analysis being limited to countries with complete data, resulting in a small number of units. Additionally, short- and long-term effects are investigated by lagging trade openness and the control variables by 1 and 5 years. Using this small sample of countries with complete data, the effects are estimated using fixed effects and first difference specifications. Additionally, the models are estimated with and without control variables to check for robustness. Due to concerns of heteroscedasticity and serial correlation, bootstrapping and clustering are employed to construct confidence intervals and to perform inference.

The resulting estimates are positive, yet small in magnitude, for both short- and long-term relationships. This may suggest that trade openness is associated with an increase in inequality. There are, however, varying results regarding the statistical significance of the estimates across specifications and inference methods. Overall, this may suggest that the effect of trade openness on the Gini index is statistically insignificant. The limited explanatory power of trade openness may indicate that more specific factors influence inequality, and thus motivate further research into the relationship between inequality and particular aspects of trade openness.

II Data

The data used for this paper is derived from the World Bank's World Development Indicators database (World Bank, 2025). This provides information for 217 economies, which includes countries, territories, and some non-independent areas which have separate economic data. For simplicity, these economies will be referred to as 'countries' through this paper.

II.1 Variables

To examine the effect of economic integration on inequality, economic integration will be represented by trade openness, and the Gini index will be used as the measure of inequality. Here trade openness is the sum of a country's imports and exports of goods and services as a percentage of their gross domestic product (GDP). This is measured annually and will be referred to as 'trade'. As for the Gini index, this measures the distribution of income within a country, and is on a scale of 0 to 100 – 0 representing perfect equality and 100 representing perfect inequality. Countries with a higher index are therefore said to have more inequality, and a positive change in the Gini index implies an increase in inequality.

Data for various controls, or factors other than trade that are likely to affect inequality, are also drawn from the same source. This includes the unemployment rate, GDP per capita, the inflation rate, and tax revenue. For consistency across countries, a modeled ILO estimate is used to measure unemployment as a percent of the total labour force. This is included to account for changes and differences in unemployment across countries, a factor that can have a great impact on a population's distribution of wealth. Similarly, GDP per capita is measured in constant 2015 United States Dollars to account for inflation and ensure a fair comparison across countries and time. Annual inflation, using consumer prices, is also of interest to control for due to inflation degrading purchasing power. It can therefore affect the standard of living, and thus possibly affect lower income groups more – putting upward pressure on income inequality. Finally, tax revenue, as a percentage of GDP, is also included to account for fiscal space, which can affect a country's ability to address inequality.

II.2 Complete Data

When pulling the data from the World Development Indicators database, not all countries have perfect or complete data. Therefore, in order to perform this analysis, countries without complete data for any of the aforementioned variables were dropped. This resulted in a small subset of 8 countries, which will be an important consideration when performing inference and interpreting results. For this subset of countries, Table 1 presents the descriptive statistics of each variable across all countries and across the entire time period. Figure 1 also presents the visual trends in these variables across time, with each of the lines representing one of the 8 countries with complete data.

Another concern lies in the reasons for which countries may lack data for certain variables. In many cases, political reasons or limitations due to infrastructure and resources may hinder a country's ability to measure and report certain indicators. These reasons, however, are also likely to affect trade openness and inequality. As a result, bias is likely to be present in the results, as the subsample of data is representative only of countries with the ability to measure or collect such information.

II.3 Temporal Dynamics

The time period of focus for this analysis is 1996 to 2019. This falls just after the creation of the World Trade Organization in 1995, when the majority of countries joined the organization which promotes economic integration, and before the COVID-19 pandemic which was a shock to trade.

Due to the nature of how the effect of trade and the various control variables on inequality may be delayed, trade and the controls are lagged. This is because inequality may be slow and difficult to change due to its complexity and connection to a variety of factors. Trade in particular may take a while to affect inequality, and the short run and long run impacts may vary greatly as countries and industries adjust. In the interest of maintaining as much data as possible, this paper examines a lag of 1 and 5 years. The lag of 1 year therefore allows for an estimate of the short-term, or immediate, effects of trade on inequality, while the 5 year lag allows for an estimate of a longer term relationship.

III Empirical Methods

Examining the effect of trade openness on inequality across countries involves looking at various countries across time. However, there are unobserved differences across both countries and time that can affect trade and inequality. Therefore given the panel data structure, a fixed effects model and a first difference model will be compared – both focusing on within country variation.

III.1 Fixed Effects Model

The fixed effects model will take into account unobserved country specific characteristics as well as time fixed effects. Country fixed effects will control for any country specific characteristics that are constant over time like geography, climate, trade agreements, and historical connections which may affect trade openness and inequality. On the other hand, time fixed effects will account for factors that change over time, but that are constant across countries, therefore accounting for factors that affect all countries. Time fixed effects may therefore capture global trends that affect trade and inequality globally, such as the Global Financial crisis in 2008. Accounting for these fixed effects ensures that the model captures the true relationship between trade openness and inequality. The fixed effects specifications will therefore take the form of Equation (1) and (2).

$$Gini_{it} = \alpha_0 + \beta_1 trade_{it} + \alpha_i + \gamma_t + u_{it} \quad (1)$$

$$Gini_{it} = \alpha_0 + \beta_1 trade_{it} + \beta' X_{it} + \alpha_i + \gamma_t + u_{it} \quad (2)$$

In both specifications, the estimate of interest will be β_1 which will capture the effect of trade openness on the Gini index. The aforementioned country fixed effects and time fixed effects are α_i and γ_t respectively. The specification of Equation (2) is slightly different, including X_{it}, a vector of control variables, as well as their respective coefficients β', which represent their effect on the Gini index. The inclusion of control variables helps to check the robustness of the estimate by controlling for other varying factors which may affect inequality. Finally, u_{it} is the error term that captures the variation in inequality that is unexplained by trade, the controls, and the fixed effects. The main assumption of the fixed effects model is that this error term is uncorrelated with trade across countries and time, which is necessary for the estimator to be consistent and for causal inference.

III.2 First Difference Model

The first difference model will also help to eliminate time-constant unobserved effects by focusing on the changes over time in trade and inequality for individual countries. This approach removes country specific characteristics that are constant over time by differencing the variables. The first difference specifications will therefore take the form of Equations (3) and (4).

$$\Delta Gini_{it} = \beta_1 \Delta trade_{it} + \Delta u_{it} \quad (3)$$

$$\Delta Gini_{it} = \beta_1 \Delta trade_{it} + \beta' \Delta X_{it} + \Delta u_{i} \quad (4)$$

In both first difference specifications, the estimate of interest will be the effect of changes in trade openness on changes in the Gini index, captured by β_1. By focusing on the change in trade, any country specific characteristics that are constant over time are differenced out. The specification of Equation (4) again includes ΔX_{it}, a vector of the changes in control variables, as well as their respective coefficients β'. These are the same time varying controls as the fixed effects model which are included for completeness, and this helps to check the robustness of the estimate by accounting for how changes in other factors may affect changes in inequality. Finally, Δu_{it} is the change in the error term that captures the additional unexplained variation in the changes of inequality. For the first difference model, the main assumption is that changes in trade are uncorrelated with changes in the error term – an assumption which is necessary for the estimator to be consistent and for causal inference.

III.3 Inference

The assumption of homoscedasticity is unlikely to hold due to concerns regarding non constant variation of the error term across countries and time. Thus the precision of the estimates can vary greatly, and this can negatively affect the reliability of the standard errors. Serial correlation is also a concern given the likelihood that the errors may be correlated over time within the same country. This arises due to this analysis looking at factors like trade openness and inequality that are persistent and only gradually change due to preexisting or previous conditions. For these reasons, the naive standard errors should not be relied upon. Instead, bootstrapping and clustering will be used to construct 95% confidence intervals to account for possible heteroscedasticity and serial correlation.

The bootstrapping method entails 10,000 resamples of the data, with replacement, producing 10,000 bootstrap estimates. Confidence intervals are then constructed based on the 2.5% and 97.5% quartiles of the bootstrap distribution. Because bootstrapping is a non-parametric approach, no assumptions are made regarding the distribution of the data. These will then be compared to the confidence intervals constructed using the clustered standard errors, however, the clustered standard errors may be less reliable due to the small number of countries with complete data. Clustering will be performed at the country level, allowing for correlation across time within the same country.

IV Results

IV.1 Fixed Effects Estimates

The fixed effects estimates looking at the short-run effect of trade openness, with a lag of 1 year, are presented in Table 2. Both specifications with and without controls suggest that a 10 percentage point increase in trade openness is associated with an increase of the Gini index of 0.19 points. With the naive standard errors, which do not account for serial correlation and heteroskedasticity, these estimates are statistically significant at the 1% and 5% levels. Given the positive estimates, this suggests that in the short-term, trade openness may slightly increase inequality due to the positive coefficients, however on the scale of the Gini index, this is a very small change in magnitude.

As for the longer-term relationship, the fixed effects estimates with a lag of 5 years are presented in Table 3. Here the impact of a 10 percentage point increase in trade openness on the Gini index is estimated to be larger, with estimated changes in the Gini of 0.37, and then 0.43 from the specification with controls. Both estimates are significant at the 1% significance level according to the naive standard errors. The positive coefficients again suggest that trade openness may contribute to more inequality in the long-run. With the longer lag, the estimate changes slightly in magnitude with the addition of controls. This would suggest negative omitted variable bias, where a control like tax revenue may help to decrease inequality. Thus controlling for other variables leads to trade openness explaining more of the variation in the Gini index. However this is a very small change in magnitude and does not make too much of a difference on the scale of the Gini index.

IV.2 First Difference Estimates

The estimates of the first difference specifications are presented in Tables 4 and 5. The estimate of the effect of trade on the Gini index, with a lag of 1 year and without controls from

Table 4, suggests that a 10 percentage point increase in trade sees an increase in the Gini index of 0.23. This is significant at the 5% level using naive standard errors, and the estimate is similar in magnitude to the corresponding fixed effects estimate in column 1 of Table 2. With a lag of 5 years, however, the model estimates a statistically insignificant change of 0.12 in the Gini index, as seen in Table 5. This is in contrast with the fixed effects estimate from column 1 of Table 3, where the estimate is 0.37 and statistically significant using the naive standard errors. Importantly, the sign of the first difference estimates are positive, suggesting a positive relationship between trade openness and the Gini index, and which is in line with the fixed effects estimates.

Column 2 of Tables 4 and 5 present the estimates from the specifications with the same controls as the fixed effects model. These effects are both statistically insignificant according to the naive standard errors, and are very small in magnitude, varying greatly from the specifications without controls. The estimate of the effect of trade on the Gini index with a lag of 5 years and controls is also negative, however it is extremely small and very close to 0 – especially on the Gini index scale of 0 - 100. One reason for the large change in the first difference estimates with the addition of controls may be the fact that these controls are time-varying controls. The first difference model specifically focuses on changes in time for a specific country, and thus controlling for these variables may reduce the variation in trade openness. Thus this may reduce the ability of trade openness to explain the variation in the Gini index, leading to very small estimates.

IV.3 Inference - Bootstrap and Clustered Confidence Intervals

Due to concerns of heteroskedasticity and serial correlation, the naive standard errors in the tables above may lead to incorrect conclusions. Thus bootstrap standard errors and clustered standard errors are computed. However, there is still a concern regarding the clustered standard errors due to the small number of clusters as a result of the limited number of units with complete data.

Figure 2 presents the bootstrap distribution for the first difference and fixed effects estimates with and without controls, and with a lag of 1 year. For all four estimates, the distributions appear to be roughly centered around the original estimate. However, the first difference estimates are more consistent, with a higher density around the original estimate. The fixed effects distributions also appear to have slightly longer left tails, with an almost bimodal distribution. This may be due to the limited number of units which means less variation in the data, and thus countries with smaller values could be over represented when resampling. A significant portion of the four distributions also fall below 0. This is reflected in Table 6, which displays the bootstrap confidence intervals, where the 95% confidence intervals for all 4 specifications contain 0. Therefore based on bootstrap inference, 0 cannot be ruled out as the true effect of trade openness on the Gini index after 1 year.

For the specifications with a lag of 5 years, Figure 3 displays the bootstrap distributions, and Table 7 contains the 95% confidence intervals. In all four cases, the bootstrap distributions are centered on the original estimates and are more consistent. While the left tails of all four distributions cross 0, a more significant portion of the distribution falls below 0 for the first difference specifications. This is again reflected in Table 7, where the 95% confidence intervals contain 0. However, the confidence intervals for the fixed effects specifications either do not contain 0, or have a lower bound that is very close to 0. This might suggest that the fixed effects estimates might be statistically significant at the 5% level, or even significant at a higher level for the specification with controls. However the fact that the majority of the confidence intervals contain 0 once again may suggest that trade openness has no effect on the Gini index after 5 years.

Clustered standard errors are also computed and presented in Table 8, for the specifications with a lag of 1 year, and in Table 9, for the specifications with a lag of 5 years. Using these standard errors, 95% confidence intervals were then computed, and these are presented in Table 10 and 11. For the specifications with a lag of 1 year, all estimates are found to be statistically insignificant at the 5% level of significance, other than the first difference estimate. This can be seen in Table 10, where the first difference is the only specification with a confidence interval that does not contain 0. The clustered inference therefore suggests that trade openness may not have an effect on the Gini index in the short-run. Table 11 suggests, however, that the estimates for all specifications, apart from the first difference model with controls, are statistically significant at the 5% level when a lag of 5 years is used. Using clustered inference, it may then be suggested that trade openness could have a statistically significant impact on the Gini index in the longer term.

A summary of the statistical significance of the estimates is included in Table 12. While the different inference methods see varying results, the naive standard errors appear to over reject the null hypothesis in comparison to the bootstrap and clustered standard errors. This is consistent with the naive standard errors failing to account for heteroskedasticity and serial correlation. Between the bootstrap and clustered standard errors, however, there is still variation in the results. The clustered inference fails to reject the null hypothesis that the true effect of trade is 0 for the 5 year fixed effects with controls, the 1 year first difference, and the 5 year first difference. For these same specifications the bootstrap inference cannot reject the null hypothesis. The only estimate for which both inference methods do reject the null hypothesis is the 5 year fixed effects estimate.

A reason for these varying results may be the few countries with complete data. Limited variation may mean certain countries are overrepresented in the bootstrapping process, and few units means fewer clusters for the clustered standard errors. This may be a more serious issue for the clustered standard errors as there is less variation between the clusters. The estimates themselves may also be less precise as a result of reduced variation in a small sample which is not truly representative of the population. Overall, it is not possible to rule out that the true effect of trade openness on the Gini index is 0 for both time frames. This is due to the varying inference results, as well as the varying results between the first difference and fixed effects estimates.

IV.4 Discussion

For the most part, the various specifications estimated that trade openness has a positive effect on the Gini index, suggesting that trade openness could increase inequality. While inference produces varying results regarding whether or not the estimates are statistically significant, the magnitude of the positive estimates is an important consideration. The estimated effect of a 10 percentage point increase in trade openness on the Gini index is on average 0.23 across all specifications. The Gini index itself ranges from 0 to 100. Therefore in a practical sense, the estimates are practically insignificant, especially considering how a 10 percentage point increase in trade openness would indicate a substantial change in trade flows relative to a country's GDP. These estimates are also likely to be biased due to the small number of units. Since complete data was needed, this analysis only used the few countries that had complete data, and thus is only representative of countries who managed to collect and record such data. It is therefore important to be critical of the fact that the reasons for why certain countries do or do not have data might also be factors that have an effect on both trade openness and the Gini index.

The models used – fixed effects and first difference – work to establish the causal effect of trade on the Gini by controlling for or eliminating time invariant country characteristics. Despite this, time-varying country characteristics may still be a concern. In the fixed effects model, which controls for time invariant country characteristics and global time specific events, the main assumption made is that the error term is uncorrelated with trade openness. This is a key assumption to ensure that the estimate is unbiased, however it may be violated by country characteristics that change over time, are connected to trade openness, and which are not controlled for by the model. The specification with controls attempts to account for some of these factors. Similarly, the first difference specification looks to eliminate time-invariant country characteristics by focusing on changes in the variables within a country. Thus the main assumption is that changes in the error term are not correlated with changes in trade openness. Once again, it is likely that this assumption may be violated by country characteristics that change over time, and which are associated with changes in trade openness.

Such country characteristics that vary over time and which are connected with trade openness may include technological and productivity advancements or improvements. These changes within a country could make their goods and services more productive, thus affecting trade openness and inequality. Similarly related, a country's comparative advantage could change over time, as countries industrialize and develop new methods and technologies. This is important for trade, possibly influencing a country's trade openness over time, as well as possibly influencing inequality depending on what jobs are created or supported by such production. Mean and/or median incomes may also change over time if countries move towards supporting their comparative advantage, possibly transitioning to be reliant on certain industries to boost trade. These changes could also have an important impact on inequality over time. Finally, the institutional capacities of countries and political environments may evolve. Such capacities and environments play an important role in their ability to form trade agreements and maintain international relations, thus influencing trade openness. These factors will also play an important role in dealing with inequality, such as with the redistribution of tax revenue to support their population.

Given the inability to reject that the true effect of trade openness on the Gini index is different from 0, trade openness appears to be a weak explanatory variable. This may be due to trade openness not being the best choice to explain the variation in the Gini index. Trade openness itself is simply the sum of a country's imports and exports as a percentage of GDP. It therefore does not have any information regarding how gains or losses from trade may be distributed among a population or industries. Trade openness also does not provide any information about the composition of this trade. Inequality could be affected differently based on whether or not this trade is driven by imports or exports, if it is trade in goods or services, or possibly be more dependent on the types of things trades. Thus, inequality may not be simply dependent on the level of trade openness, but more the specific characteristics of this economic activity and integration.

V Conclusion

In using trade openness and the Gini index to answer the question of the effect of economic integration on inequality, the varying inference results and practically insignificant estimates may suggest that trade openness does not have an effect on the Gini index. This comes after comparing the fixed effects estimates to the first difference estimates, and exploring both short- and long-term effects. In the end, bootstrapping and clustering at the country level to account for heteroskedasticity and serial correlation produce 95% confidence intervals that contain 0, or are very close to 0. Regardless of significance, however, the estimates are negligible on the scale of the Gini index.

These results suggest that trade openness has a practically and statistically insignificant effect on inequality, and thus has very little explanatory power. It is also important to keep in mind that these results are likely to be biased due to the small number of countries with complete data. Time-varying country characteristics that may be unobserved or difficult to control for are also a concern, as they may be correlated with trade openness and inequality, and thus violate the main assumptions of the models. Both the first difference and fixed effects models control for or eliminate time-invariant country characteristics or global time-trends, however time-varying country characteristics may still impact the consistency of the estimate.

If trade openness alone does not have an effect on inequality, then this may motivate further research into the impact the composition of trade has on inequality – such as if increased imports or exports are found to have varying effects. Additional focus on specific trade policies may also serve to be more fruitful in estimating their specific impacts on society, as opposed to an aggregate measure like trade openness, which may lack a specificity needed to capture effects on societal conditions. Ultimately, trade openness is shown to have very little explanatory power with respect to the Gini index, and changes in inequality may instead be attributed to more complex dynamics, requiring additional research to explore connections between economic integration and inequality.

References

World Bank. (2024). World Development Indicators. World Bank Group. Retrieved March 6, 2025, from https://databank.worldbank.org/source/world-development-indicators

Table 1: Descriptive Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Trade (% GDP)	192	90.91	80.19	22.29	382.35
Gini Index	192	36.18	5.83	28.00	51.80
Unemployment Rate	192	7.78	3.58	1.80	20.71
GDP per Capita	192	40,090.01	28,028.83	1,187.12	112,417.90
Inflation Rate	192	3.20	4.16	-0.94	39.36
Tax Revenue (% GDP)	192	17.31	6.12	7.88	26.64

Figure 1: Variables Over Time

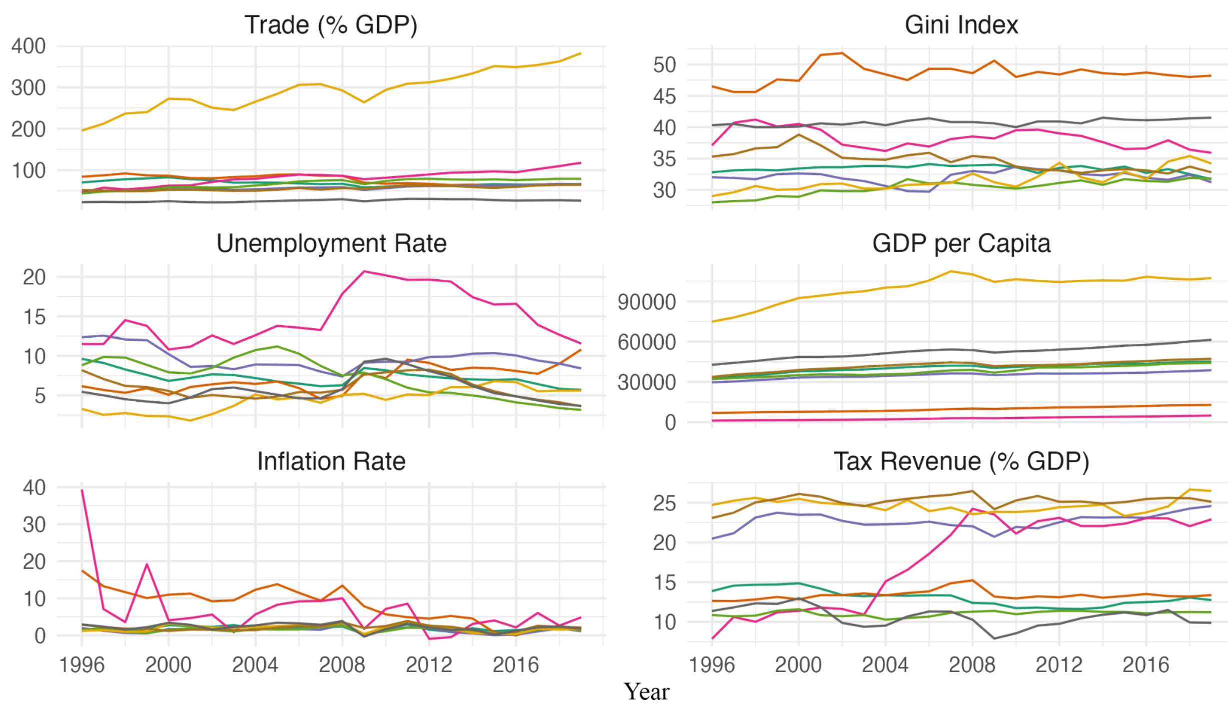

Table 2: Fixed Effects Regression Results (1 Year Lag)

	Dependent variable:	
	Gini	Gini with Controls
	(1)	(2)
Trade (% GDP)	0.019***	0.019**
	(0.006)	(0.008)
Gini Index		0.058
		(0.065)
Unemployment Rate		0.00001
		(0.00005)
GDP per Capita		0.042
		(0.034)
Inflation Rate		-0.148**
		(0.058)
Observations	184	184
R^2	0.064	0.144
Adjusted R^2	-0.119	-0.051
F Statistic	10.517*** (df = 1; 153)	5.028*** (df = 5; 149)

Note: Naive standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 3: Fixed Effects Regression Results (5 Year Lag)

	Dependent variable:	
	Gini	Gini with Controls
	(1)	(2)
Trade (% GDP)	0.037***	0.043***
	(0.006)	(0.008)
Gini Index		-0.126**
		(0.056)
Unemployment Rate		-0.00002
		(0.00004)
GDP per Capita		0.027
		(0.027)
Inflation Rate		-0.032
		(0.051)
Observations	152	152
R^2	0.241	0.312
Adjusted R^2	0.083	0.141
F Statistic	39.675*** (df = 1; 125)	10.957*** (df = 5; 121)

Note: Naive standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 4: First Difference Regression Results (1 Year Lag)

	Dependent variable:	
	Gini	Gini with Controls
	(1)	(2)
Trade (% GDP)	0.023**	0.008
	(0.010)	(0.012)
Gini Index		0.007
		(0.074)
Unemployment Rate		0.0002**
		(0.0001)
GDP per Capita		0.014
		(0.020)
Inflation Rate		0.148*
		(0.082)
Tax Revenue (% GDP)	-0.026	-0.103
	(0.072)	(0.078)
Observations	176	176
R^2	0.028	0.078
Adjusted R^2	0.022	0.051
F Statistic	5.005** (df = 1; 174)	2.878** (df = 5; 170)

Note: Naive standard errors in parentheses.

*** p < 0.01, ** p <0.05, * p <0.1

Table 5: First Difference Regression Results (5 Year Lag)

	Dependent variable:	
	Gini	Gini with Controls
	(1)	(2)
Trade (% GDP)	0.012	-0.0001
	(0.010)	(0.013)
Gini Index		-0.023
		(0.078)
Unemployment Rate		0.0001*
		(0.0001)
GDP per Capita		0.018
		(0.020)
Inflation Rate		-0.076
		(0.086)
Tax Revenue (% GDP)	-0.077	-0.121
	(0.077)	(0.086)
Observations	144	144
R^2	0.009	0.047
Adjusted R^2	0.002	0.013
F Statistic	1.289 (df = 1; 142)	1.376 (df = 5; 138)

Note: Naive standard errors in parentheses.

*** p < 0.01, ** p <0.05, * p <0.1

Figure 2: Bootstrap Distribution of Estimates (1 Year Lag)

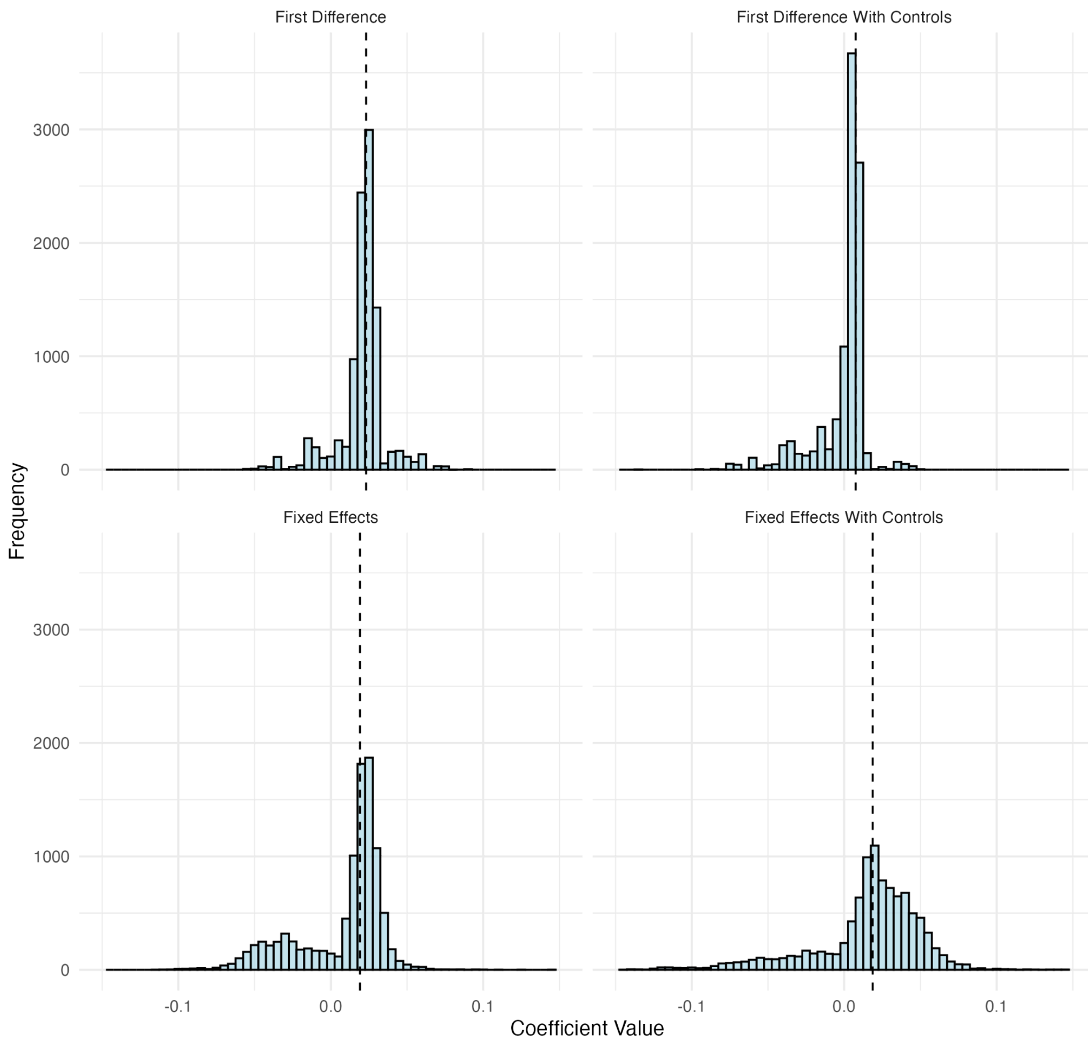

Table 6: Bootstrap Confidence Intervals (1 Year Lag)

Model	Lower Bound	Upper Bound
Fixed Effects	-0.058	0.041
Fixed Effects With Controls	-0.077	0.068
First Difference	-0.017	0.054
First Difference With Controls	-0.049	0.016

Table 7: Bootstrap Confidence Intervals (5 Year Lag)

Model	Lower Bound	Upper Bound
Fixed Effects	0.012	0.052
Fixed Effects With Controls	-0.0001	0.072
First Difference	-0.019	0.017
First Difference With Controls	-0.018	0.035

Figure 3: Bootstrap Distribution of Estimates (5 Year Lag)

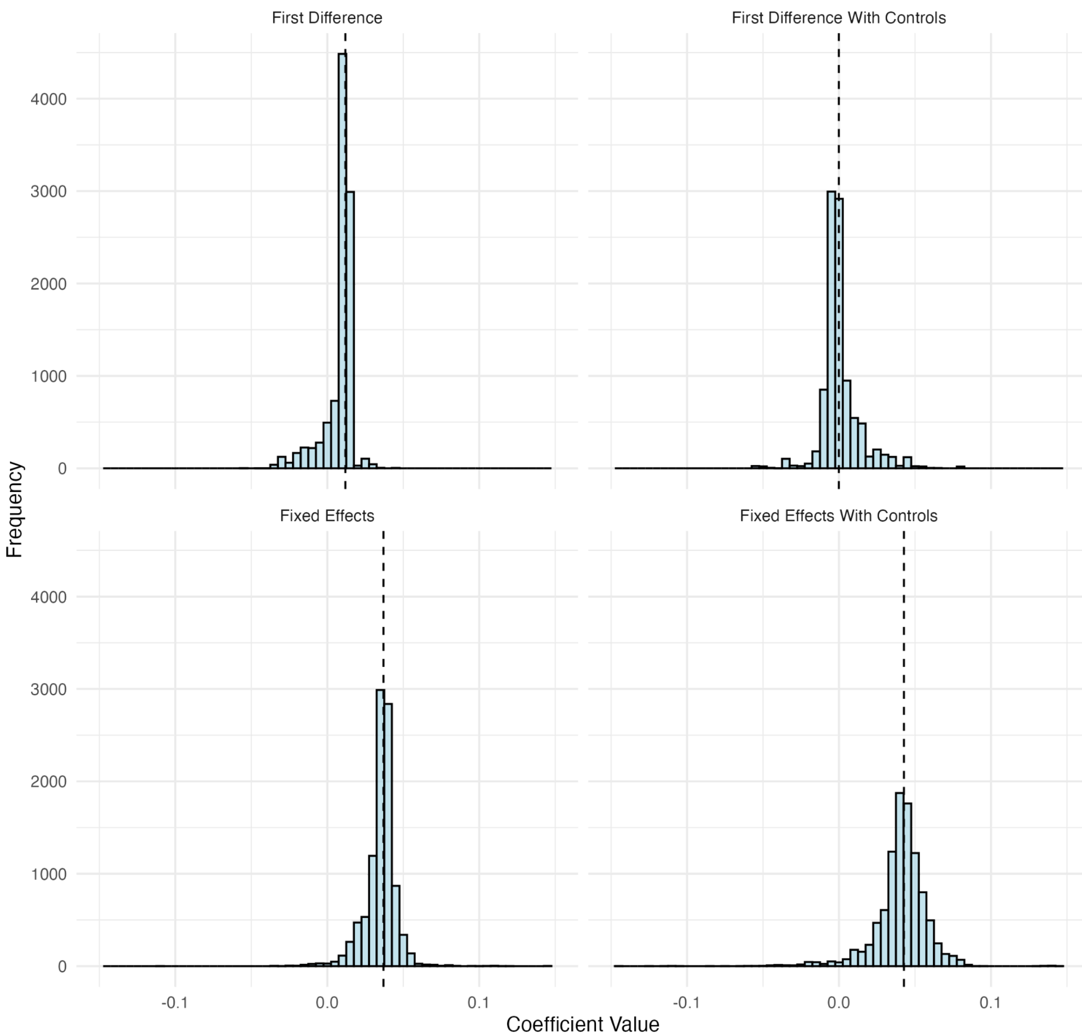

Table 8: Clustered Standard Errors (1 Year Lag)

Model	Standard Error
Fixed Effects	0.012
Fixed Effects With Controls	0.015
First Difference	0.007
First Difference With Controls	0.008

Table 9: Clustered Standard Errors (5 Year Lag)

Model	Standard Error
Fixed Effects	0.004
Fixed Effects With Controls	0.007
First Difference	0.004
First Difference With Controls	0.008

Table 10: Clustered Confidence Intervals (1 Year Lag)

Model	Lower Bound	Upper Bound
Fixed Effects	-0.004	0.042
Fixed Effects With Controls	-0.011	0.048
First Difference	0.010	0.036
First Difference With Controls	-0.009	0.024

Note: The confidence intervals are calculated using the standard errors presented in Table 8 and follow the standard equation of $\hat{\beta} \pm 1.96 \cdot SE(\hat{\beta})$.

Table 11: Clustered Confidence Intervals (5 Year Lag)

Model	Lower Bound	Upper Bound
Fixed Effects	0.029	0.045
Fixed Effects With Controls	0.030	0.055
First Difference	0.004	0.020
First Difference With Controls	-0.015	0.015

Note: The confidence intervals are calculated using the standard errors presented in Table 9 and follow the standard equation of $\hat{\beta} \pm 1.96 \cdot SE(\hat{\beta})$.

Table 12: Statistical Significance Summary

Model	Naïve Standard Errors	Bootstrap Standard Errors	Clustered Standard Errors
Fixed Effects (1 Year Lag)	Yes	No	No
Fixed Effects with Controls (1 Year Lag)	Yes	No	No
Fixed Effects (5 Year Lag)	Yes	Yes	Yes
Fixed Effects with Controls (5 Year Lag)	Yes	No	Yes
First Difference (1 Year Lag)	Yes	No	Yes
First Difference with Controls (1 Year Lag)	No	No	No
First Difference (5 Year Lag)	No	No	Yes
First Difference with Controls (5 Year Lag)	No	No	No

Note: "Yes" indicates that the coefficient on trade openness is statistically significant at the 5% level based on the 95% confidence intervals.